Excel Product Sales Data Exploration

Excel File: Excel Product Sales Data Exploration

Task Description:

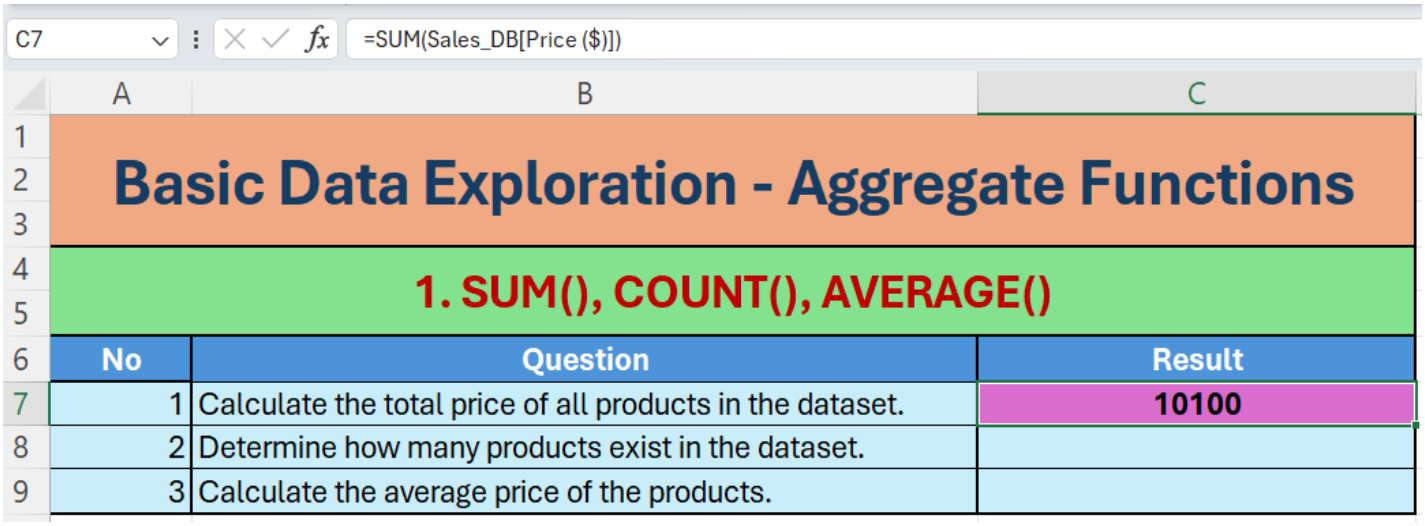
Basic Data Exploration:

- **Aggregate Functions** → *SUM, COUNTA, AVERAGE, MIN, MAX*
- **Logical Function** → *IF*
- **Conditional Functions** → *SUMIF, COUNTIF*
- **Text Functions** → *LEFT, RIGHT, MID*

Aggregate Functions:

1. Calculate the total price of all products in the dataset:

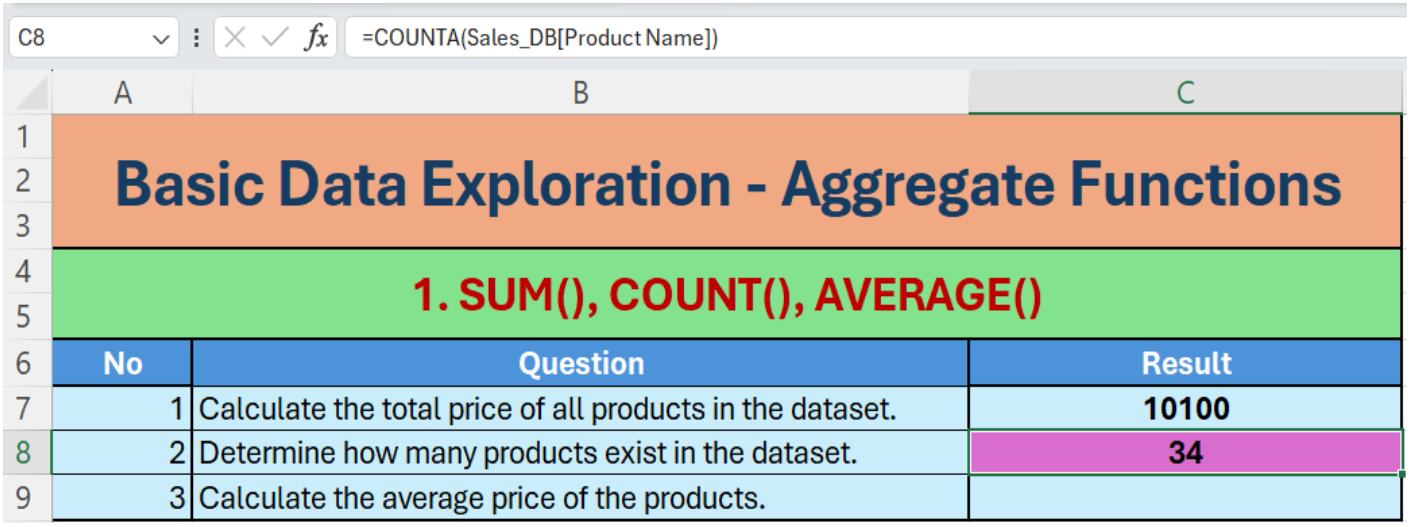
   Function: *=SUM(Sales_DB[Price ($)])*



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2. Determine how many products exist in the dataset:

Function: *=COUNTA(Sales_DB[Product Name])*

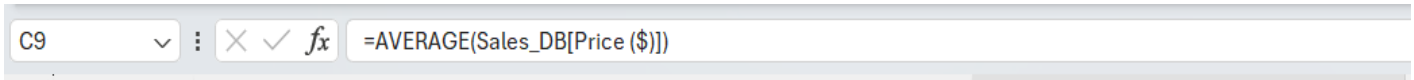


	A	B	C
1			
2		**Basic Data Exploration - Aggregate Functions**	
3			
4		**1. SUM(), COUNT(), AVERAGE()**	
5			
6	**No**	**Question**	**Result**
7	1	Calculate the total price of all products in the dataset.	**10100**
8	2	Determine how many products exist in the dataset.	**34**
9	3	Calculate the average price of the products.	

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3. Calculate the average price of the products:

Function: *=AVERAGE(Sales_DB[Price ($)])*

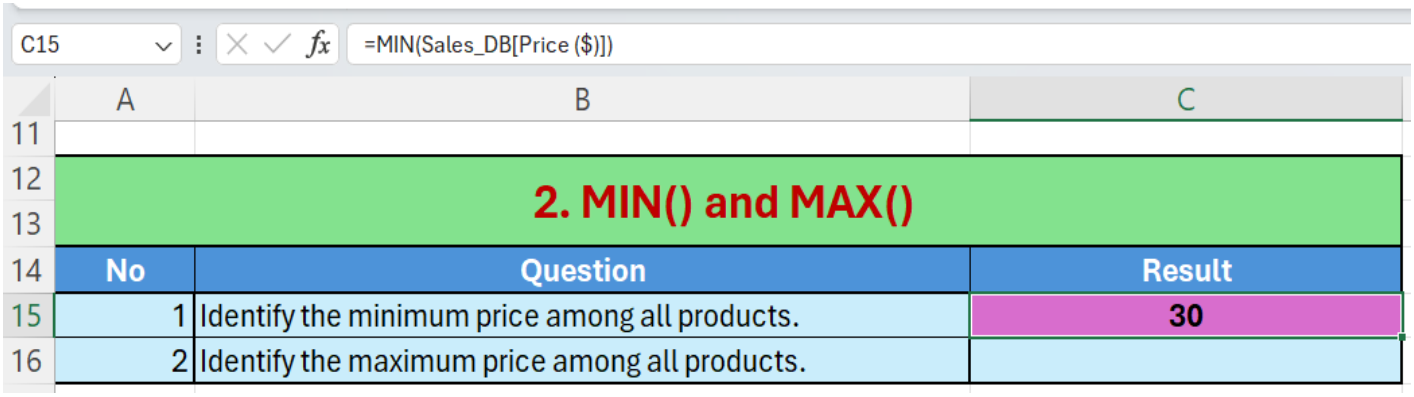


	A	B	C
1			
2		**Basic Data Exploration - Aggregate Functions**	
3			
4		**1. SUM(), COUNT(), AVERAGE()**	
5			
6	**No**	**Question**	**Result**
7	1	Calculate the total price of all products in the dataset.	**10100**
8	2	Determine how many products exist in the dataset.	**34**
9	3	Calculate the average price of the products.	**297.0588235**

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4. Identify the minimum price among all products:

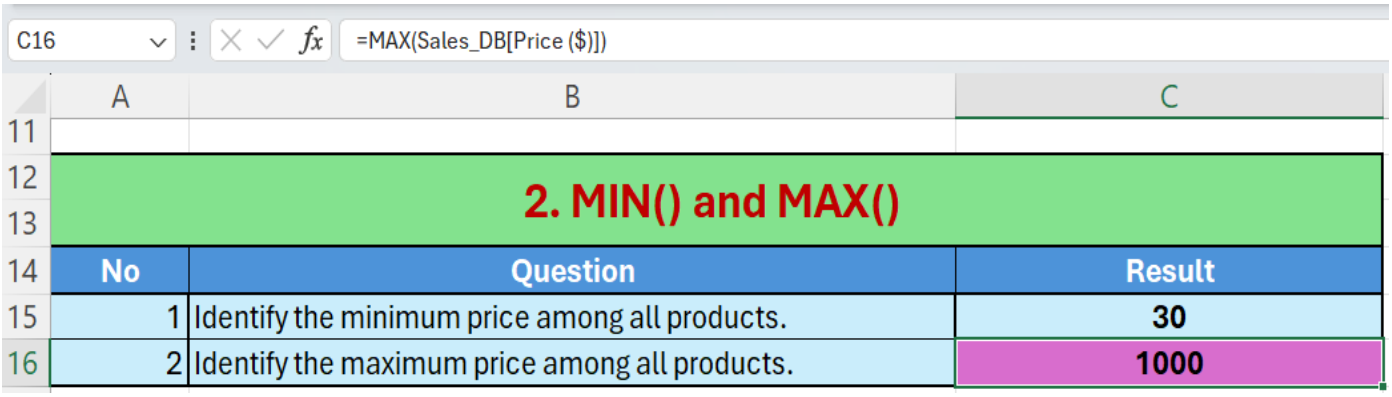
Function: *=MIN(Sales_DB[Price ($)])*



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5. Identify the maximum price among all products:

Function: *=MAX(Sales_DB[Price ($)])*



Logical Function:

1. Create a new column named *Price Range*
   Use the IF function to categorize products based on their price:
   - If Price ≥ $500 → classify as **High Price**
   - If Price < $500 → classify as **Standard Price**

Function:

=IF([@[Price ($)]]>=500,"High Price",“Standard Price")

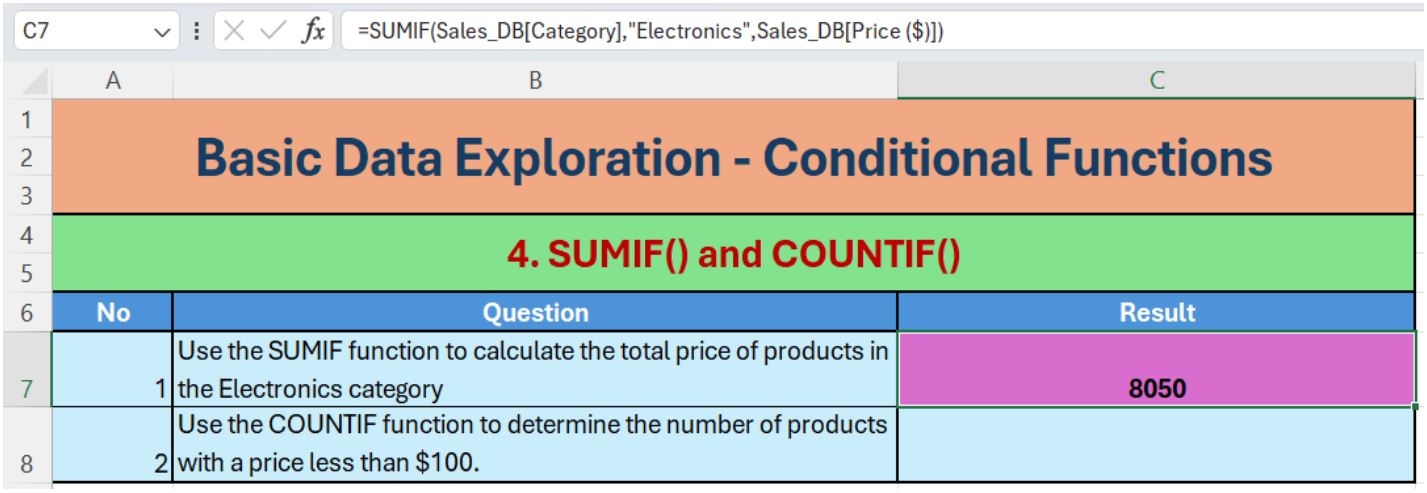
G2 | =IF([@[Price ($)]]>=500,"High Price","Standard Price")

	A	B	C	D	E	F	G
1	Product ID	Product Name	Brand Name	Category	Price ($)	Quantity	Price Range
2	28-JAN-US	Laptop	Dell	Electronics	1000	30	High Price
3	15-FEB-US	Sneakers	Nike	Fashion	80	15	Standard Price
4	03-MAR-US	Coffee Maker	Keurig	Kitchen	130	40	Standard Price
5	11-APR-US	Smartphone	Samsung	Electronics	900	25	High Price
6	22-MAY-US	Backpack	North Face	Outdoor	70	20	Standard Price
7	07-JUN-UK	Headphones	Sony	Electronics	200	45	Standard Price
8	19-JUL-UK	T-shirt	Adidas	Fashion	30	5	Standard Price
9	23-AUG-UK	Blender	Ninja	Kitchen	90	35	Standard Price
10	05-SEP-UK	Tablet	Apple	Electronics	500	50	High Price
11	14-OCT-UK	Hiking Boots	Timberland	Outdoor	130	10	Standard Price
12	17-JUN-IN	Laptop	HP	Electronics	950	25	High Price
13	25-NOV-AU	Sneakers	Adidas	Fashion	90	40	Standard Price
14	08-DEC-DE	Coffee Maker	Nespresso	Kitchen	120	35	Standard Price
15	18-FEB-CA	Smartwatch	Fitbit	Electronics	150	15	Standard Price
16	16-APR-ES	Headphones	Bose	Electronics	250	20	Standard Price
17	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	Standard Price
18	20-AUG-CN	Smartwatch	Huawei	Electronics	160	15	Standard Price
19	27-JAN-IT	Laptop	Asus	Electronics	980	10	High Price
20	01-MAR-UK	Sunglasses	Oakley	Fashion	150	15	Standard Price
21	14-AUG-US	Camping Tent	Coleman	Outdoor	200	10	Standard Price
22	14-MAY-RU	Camera	Nikon	Electronics	700	50	High Price
23	09-JAN-CA	Microwave	Panasonic	Kitchen	80	20	Standard Price
24	19-JUL-BR	Fitness Tracker	Xiaomi	Electronics	150	30	Standard Price
25	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	Standard Price
26	29-SEP-CA	Smartphone	Google	Electronics	800	45	High Price
27	03-JUN-CA	Sunglasses	Ray-Ban	Fashion	130	25	Standard Price
28	11-JUL-CA	Blender	Vitamix	Kitchen	400	40	Standard Price
29	16-APR-ES	Headphones	Bose	Electronics	230	20	Standard Price
30	07-MAR-CA	Dress	Zara	Fashion	60	30	Standard Price
31	13-APR-CA	Toaster	Hamilton	Kitchen	40	10	Standard Price
32	24-MAY-CA	Fitness Tracker	Garmin	Electronics	130	5	Standard Price
33	02-DEC-CA	Jeans	Levi's	Fashion	50	50	Standard Price
34	17-JUN-IN	Laptop	HP	Electronics	950	25	High Price
35	09-JUL-FR	Watch	Casio	Accessories	100	20	Standard Price

Conditional Functions:

1. Use the SUMIF function to calculate the total price of products in the Electronics category:

   Function:
   =SUMIF(Sales_DB[Category],"Electronics",Sales_DB[Price ($)])

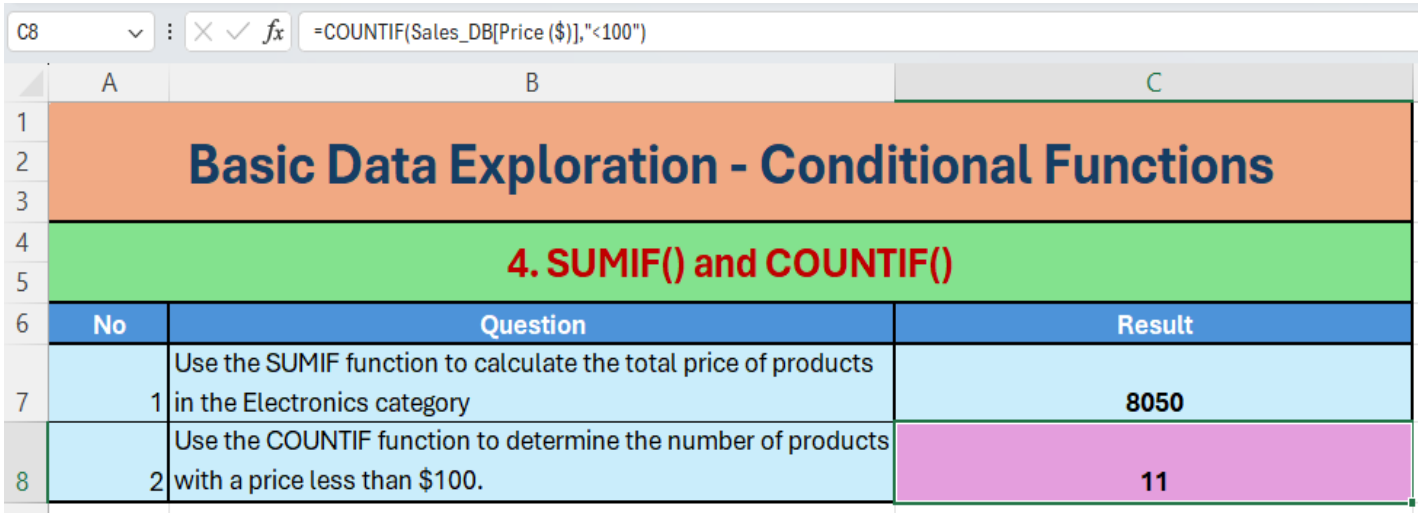


	A	B	C
1			
2		**Basic Data Exploration - Conditional Functions**	
3			
4		**4. SUMIF() and COUNTIF()**	
5			
6	**No**	**Question**	**Result**
7	1	Use the SUMIF function to calculate the total price of products in the Electronics category	**8050**
8	2	Use the COUNTIF function to determine the number of products with a price less than $100.	

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2. Use the COUNTIF function to determine the number of products with a price less than $100:

   Function: ***=COUNTIF(Sales_DB[Price ($)],"<100")***

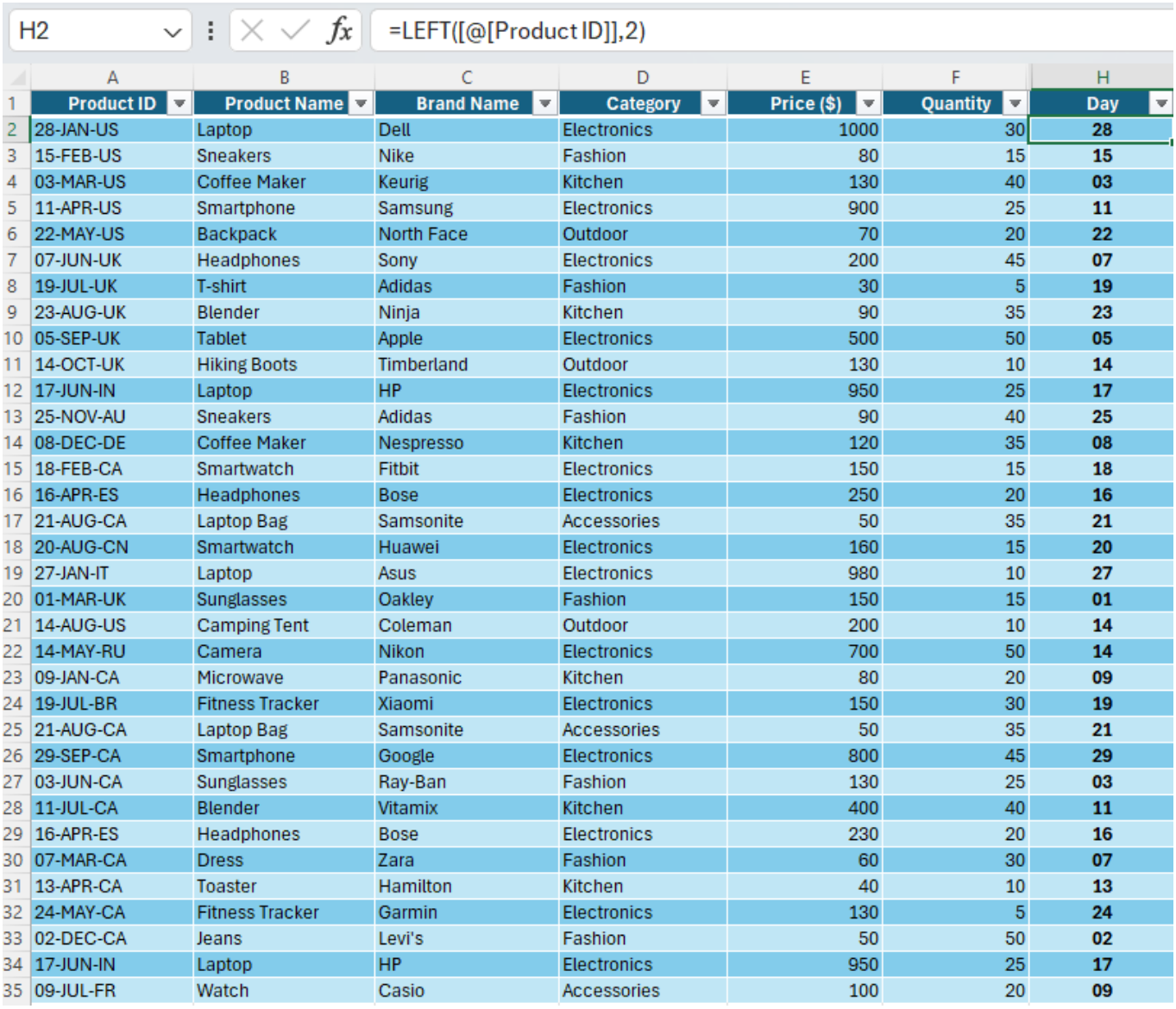


	A	B	C
1			
2		**Basic Data Exploration - Conditional Functions**	
3			
4		**4. SUMIF() and COUNTIF()**	
5			
6	**No**	**Question**	**Result**
7	1	Use the SUMIF function to calculate the total price of products in the Electronics category	**8050**
8	2	Use the COUNTIF function to determine the number of products with a price less than $100.	**11**

Text Functions:

1. Create a new column named ***"Day"*** with the first 2 characters of each 'Product ID' using the LEFT function:

   Function: *=LEFT([@[Product ID]],2)*



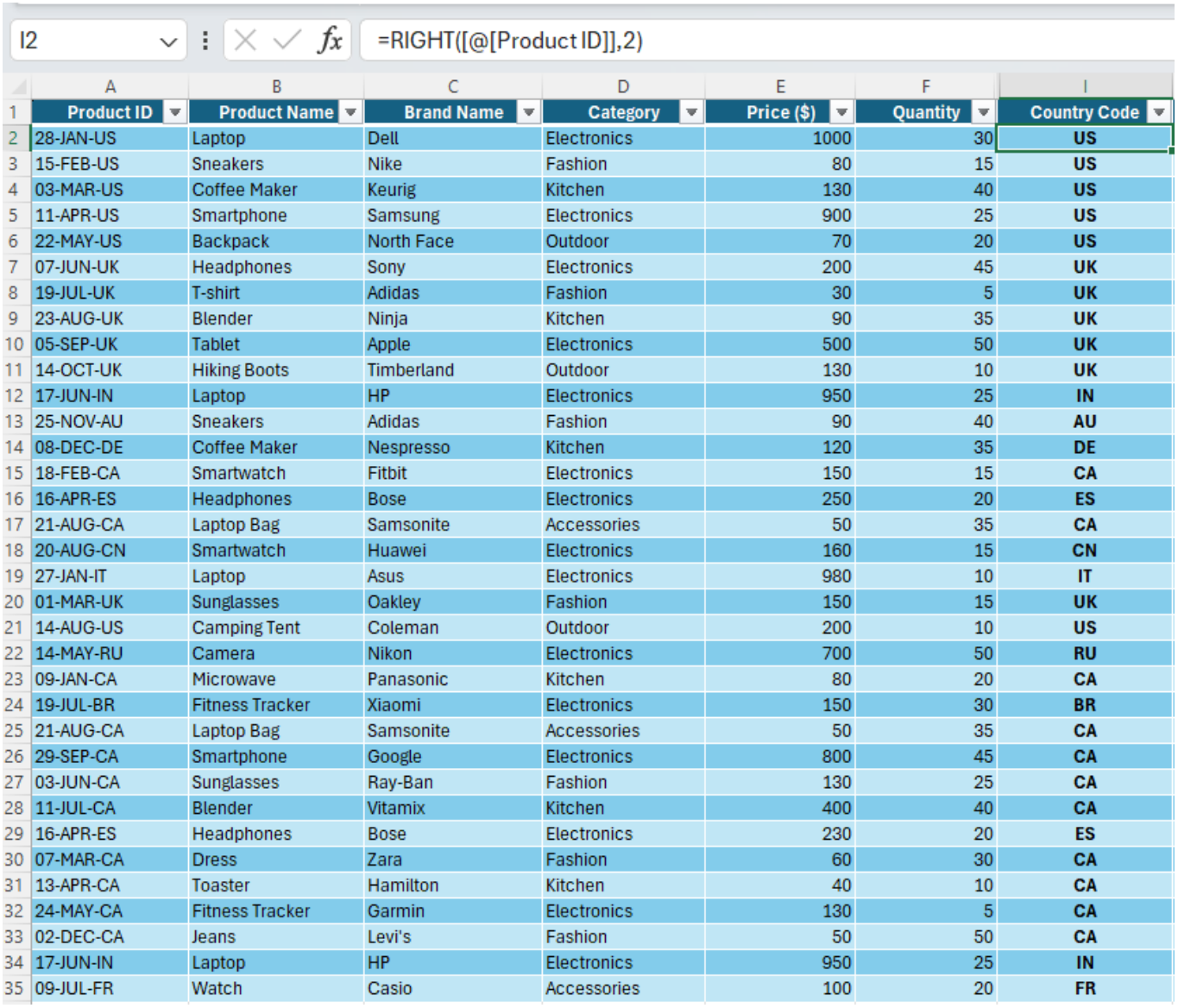
H2 | =LEFT([@[Product ID]],2)

	A	B	C	D	E	F	H
1	Product ID	Product Name	Brand Name	Category	Price ($)	Quantity	Day
2	28-JAN-US	Laptop	Dell	Electronics	1000	30	28
3	15-FEB-US	Sneakers	Nike	Fashion	80	15	15
4	03-MAR-US	Coffee Maker	Keurig	Kitchen	130	40	03
5	11-APR-US	Smartphone	Samsung	Electronics	900	25	11
6	22-MAY-US	Backpack	North Face	Outdoor	70	20	22
7	07-JUN-UK	Headphones	Sony	Electronics	200	45	07
8	19-JUL-UK	T-shirt	Adidas	Fashion	30	5	19
9	23-AUG-UK	Blender	Ninja	Kitchen	90	35	23
10	05-SEP-UK	Tablet	Apple	Electronics	500	50	05
11	14-OCT-UK	Hiking Boots	Timberland	Outdoor	130	10	14
12	17-JUN-IN	Laptop	HP	Electronics	950	25	17
13	25-NOV-AU	Sneakers	Adidas	Fashion	90	40	25
14	08-DEC-DE	Coffee Maker	Nespresso	Kitchen	120	35	08
15	18-FEB-CA	Smartwatch	Fitbit	Electronics	150	15	18
16	16-APR-ES	Headphones	Bose	Electronics	250	20	16
17	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	21
18	20-AUG-CN	Smartwatch	Huawei	Electronics	160	15	20
19	27-JAN-IT	Laptop	Asus	Electronics	980	10	27
20	01-MAR-UK	Sunglasses	Oakley	Fashion	150	15	01
21	14-AUG-US	Camping Tent	Coleman	Outdoor	200	10	14
22	14-MAY-RU	Camera	Nikon	Electronics	700	50	14
23	09-JAN-CA	Microwave	Panasonic	Kitchen	80	20	09
24	19-JUL-BR	Fitness Tracker	Xiaomi	Electronics	150	30	19
25	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	21
26	29-SEP-CA	Smartphone	Google	Electronics	800	45	29
27	03-JUN-CA	Sunglasses	Ray-Ban	Fashion	130	25	03
28	11-JUL-CA	Blender	Vitamix	Kitchen	400	40	11
29	16-APR-ES	Headphones	Bose	Electronics	230	20	16
30	07-MAR-CA	Dress	Zara	Fashion	60	30	07
31	13-APR-CA	Toaster	Hamilton	Kitchen	40	10	13
32	24-MAY-CA	Fitness Tracker	Garmin	Electronics	130	5	24
33	02-DEC-CA	Jeans	Levi's	Fashion	50	50	02
34	17-JUN-IN	Laptop	HP	Electronics	950	25	17
35	09-JUL-FR	Watch	Casio	Accessories	100	20	09

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2. Create a new column named ***"Country Code"*** by extracting the last 2 characters from the 'Product ID' column using the RIGHT function:

Function: ***=RIGHT([@[Product ID]],2)***



I2 | =RIGHT([@[Product ID]],2)

	A	B	C	D	E	F	I
1	Product ID	Product Name	Brand Name	Category	Price ($)	Quantity	Country Code
2	28-JAN-US	Laptop	Dell	Electronics	1000	30	US
3	15-FEB-US	Sneakers	Nike	Fashion	80	15	US
4	03-MAR-US	Coffee Maker	Keurig	Kitchen	130	40	US
5	11-APR-US	Smartphone	Samsung	Electronics	900	25	US
6	22-MAY-US	Backpack	North Face	Outdoor	70	20	US
7	07-JUN-UK	Headphones	Sony	Electronics	200	45	UK
8	19-JUL-UK	T-shirt	Adidas	Fashion	30	5	UK
9	23-AUG-UK	Blender	Ninja	Kitchen	90	35	UK
10	05-SEP-UK	Tablet	Apple	Electronics	500	50	UK
11	14-OCT-UK	Hiking Boots	Timberland	Outdoor	130	10	UK
12	17-JUN-IN	Laptop	HP	Electronics	950	25	IN
13	25-NOV-AU	Sneakers	Adidas	Fashion	90	40	AU
14	08-DEC-DE	Coffee Maker	Nespresso	Kitchen	120	35	DE
15	18-FEB-CA	Smartwatch	Fitbit	Electronics	150	15	CA
16	16-APR-ES	Headphones	Bose	Electronics	250	20	ES
17	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	CA
18	20-AUG-CN	Smartwatch	Huawei	Electronics	160	15	CN
19	27-JAN-IT	Laptop	Asus	Electronics	980	10	IT
20	01-MAR-UK	Sunglasses	Oakley	Fashion	150	15	UK
21	14-AUG-US	Camping Tent	Coleman	Outdoor	200	10	US
22	14-MAY-RU	Camera	Nikon	Electronics	700	50	RU
23	09-JAN-CA	Microwave	Panasonic	Kitchen	80	20	CA
24	19-JUL-BR	Fitness Tracker	Xiaomi	Electronics	150	30	BR
25	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	CA
26	29-SEP-CA	Smartphone	Google	Electronics	800	45	CA
27	03-JUN-CA	Sunglasses	Ray-Ban	Fashion	130	25	CA
28	11-JUL-CA	Blender	Vitamix	Kitchen	400	40	CA
29	16-APR-ES	Headphones	Bose	Electronics	230	20	ES
30	07-MAR-CA	Dress	Zara	Fashion	60	30	CA
31	13-APR-CA	Toaster	Hamilton	Kitchen	40	10	CA
32	24-MAY-CA	Fitness Tracker	Garmin	Electronics	130	5	CA
33	02-DEC-CA	Jeans	Levi's	Fashion	50	50	CA
34	17-JUN-IN	Laptop	HP	Electronics	950	25	IN
35	09-JUL-FR	Watch	Casio	Accessories	100	20	FR

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3. Create a new column named ***"Month"*** by extracting the 4th to 6th characters from the 'Product ID' column using the MID function:

Function: *=MID([@[Product ID]],4,3)*

J2 | =MID([@[Product ID]],4,3)

	A	B	C	D	E	F	J
1	Product ID	Product Name	Brand Name	Category	Price ($)	Quantity	Month
2	28-JAN-US	Laptop	Dell	Electronics	1000	30	JAN
3	15-FEB-US	Sneakers	Nike	Fashion	80	15	FEB
4	03-MAR-US	Coffee Maker	Keurig	Kitchen	130	40	MAR
5	11-APR-US	Smartphone	Samsung	Electronics	900	25	APR
6	22-MAY-US	Backpack	North Face	Outdoor	70	20	MAY
7	07-JUN-UK	Headphones	Sony	Electronics	200	45	JUN
8	19-JUL-UK	T-shirt	Adidas	Fashion	30	5	JUL
9	23-AUG-UK	Blender	Ninja	Kitchen	90	35	AUG
10	05-SEP-UK	Tablet	Apple	Electronics	500	50	SEP
11	14-OCT-UK	Hiking Boots	Timberland	Outdoor	130	10	OCT
12	17-JUN-IN	Laptop	HP	Electronics	950	25	JUN
13	25-NOV-AU	Sneakers	Adidas	Fashion	90	40	NOV
14	08-DEC-DE	Coffee Maker	Nespresso	Kitchen	120	35	DEC
15	18-FEB-CA	Smartwatch	Fitbit	Electronics	150	15	FEB
16	16-APR-ES	Headphones	Bose	Electronics	250	20	APR
17	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	AUG
18	20-AUG-CN	Smartwatch	Huawei	Electronics	160	15	AUG
19	27-JAN-IT	Laptop	Asus	Electronics	980	10	JAN
20	01-MAR-UK	Sunglasses	Oakley	Fashion	150	15	MAR
21	14-AUG-US	Camping Tent	Coleman	Outdoor	200	10	AUG
22	14-MAY-RU	Camera	Nikon	Electronics	700	50	MAY
23	09-JAN-CA	Microwave	Panasonic	Kitchen	80	20	JAN
24	19-JUL-BR	Fitness Tracker	Xiaomi	Electronics	150	30	JUL
25	21-AUG-CA	Laptop Bag	Samsonite	Accessories	50	35	AUG
26	29-SEP-CA	Smartphone	Google	Electronics	800	45	SEP
27	03-JUN-CA	Sunglasses	Ray-Ban	Fashion	130	25	JUN
28	11-JUL-CA	Blender	Vitamix	Kitchen	400	40	JUL
29	16-APR-ES	Headphones	Bose	Electronics	230	20	APR
30	07-MAR-CA	Dress	Zara	Fashion	60	30	MAR
31	13-APR-CA	Toaster	Hamilton	Kitchen	40	10	APR
32	24-MAY-CA	Fitness Tracker	Garmin	Electronics	130	5	MAY
33	02-DEC-CA	Jeans	Levi's	Fashion	50	50	DEC
34	17-JUN-IN	Laptop	HP	Electronics	950	25	JUN
35	09-JUL-FR	Watch	Casio	Accessories	100	20	JUL
